Joseph Tran

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Education

University of Nevada, Reno School of Medicine — Reno, NV
M.D./Ph.D. Candidate in Integrative Neuroscience — *Aug. 2023 – 2030, anticipated*

Stanford University — Stanford, CA
Bachelor of Science in Computer Science — *Aug. 2018 – March 2023*

Honors and Awards

2026: AWE Reality Hack Finalist, MIT Reality Hack x Snap
- Top 4 of 20+ competing teams

2025: Jerry and Janet Zebrack Family Endowed Scholarship for Humanism, University of Nevada, Reno School of Medicine (UNR SOM)
- Nominated by classmates and chosen by an official selection committee for demonstrating humility, respect, and humanistic values in patient care and teamwork
- Recognized with a $4,500 prize

2024: Student Outreach Clinic Summer Project Grant, UNR SOM
- Awarded $500 to pursue a research project through the Student Outreach Clinic

2023: 1st Place Poster Presentation, Inaugural Wolf Pack Pediatric Symposium
2023: AOA Grant, AOA
- Awarded $1,000 to create winter kits for homeless individuals throughout Reno.

Research

Graduate Research Assistant — Oct. 2023 – Current
Human Machine Perception Lab, UNR; Advisor: Alireza Tavakkoli, Ph.D. — *Reno, NV*
- Developing suite of visual ophthalmic tests in virtual reality to better understand visual perception.
- Implementing Vision Suite of tests in Student Outreach Clinic for feasibility study. Included IRB submission.
- Collaborating with NASA to develop VR platform for monitoring eye health microgravity to better understand Spaceflight-Associated Neuro-Ocular Syndrome (SANS)

Research Assistant — Oct. 2023 – Current
Stanford Byers Eye Institute; Advisor: Sophia Wang, M.D. — *Stanford, CA*
- Conducted research on glaucoma detection in OCT images using RETFound, a deep learning model.
- Preprocessed and analyzed medical imaging data to train and evaluate model performance.
- Assisted in developing and optimizing machine learning pipelines for automated glaucoma diagnosis.

Stanford Computer Science Honors Thesis Research — Aug. 2021 – Mar. 2023
Visualization Lab, Wu Tsai Institute — *Stanford, CA*
- Designing virtual reality Disparometer application to test whether VR headsets are a viable method for conducting vision research on patients affected with strabismus.
- VR Application will be tested on clinical patients and results will be presented to panel of Stanford Professors.

Software Engineer — Jun. 2021 – Sep. 2021
BAYER Pharmaceuticals — *St. Louis, MO (remote)*
- Developed virtual reality training program to help Bayer operators learn complex equipment terminology.
- Reduced polycount by 90% of 3D modeled equipment.

Undergraduate Research Assistant — June 2019 – June 2020
Incubator for Medical Mixed Reality (IMMERS) Lab — *Stanford, CA*
- Expanded upon traditional methods of medicine and healthcare by leveraging mixed reality technology.
- Developed an augmented reality (AR) application using the Microsoft Hololens 1, visualizing scattered x-ray to improve education and awareness of scattered radiation in interventional suites. Resulted in a poster presentation.
- Creating an AR application to aid Transcranial Magnetic Stimulation (TMS) technicians by visualizing EEG fields of stimulated brain areas using precomputed MRI patient data and Matlab algorithms.

Publications

[1] Joseph Tran et al. "Abstract HUP2: Geographical Trends in Cerebrovascular Disease Mortality in the United States, 1999-2020". In: *Stroke* 56 (Suppl_1 Feb. 2025). Publisher: American Heart Association, AHUP2–AHUP2. DOI: 10.1161/str.56.suppl_1.HUP2. URL: https://www.ahajournals.org/doi/10.1161/str.56.suppl_1.HUP2.

Currently Under Review

Tran J et al. (2026) "Evaluating RETFound: A Foundation Model for Glaucoma Detection Evaluated using Macular Optical Coherence Tomography Images"

Maturi J, Tran J et al. (2026) "Early Utilization Patterns and Adverse Outcomes of Anti-Complement Therapy in Geographic Atrophy in the Clinical Setting"

Presentations

Tran J. *Designing a Virtual-Reality Neuro-Visual Assessment Platform for Spaceflight-Associated Neuro-Ocular Syndrome*. Speaker Presentation at Microgravity to Medicine Symposium, International VR Healthcare Association (IVRHA); Orlando, Florida; October 27, 2026.

Tran J, Ravindranath R, Chang R, Wang S. *Evaluating RetFound: A Foundation Model for Glaucoma Detection Using Macular OCT Images*. Poster Presentation for ARVO Annual Meeting; Salt Lake City, Utah; May 5, 2025.

Maturi J, **Tran J**, Boucher N, Winsor J, Scott A. *Utilization Patterns and Adverse Outcomes of Anti-Complement Therapy in Geographic Atrophy in the Clinical Setting*. Poster Presentation at for ARVO Annual Meeting. Poster awarded MIT Outstanding Poster for section (presented by Maturi J); Salt Lake City, Utah; May 5, 2025.

Tran J, Pasia LG, Flores D. *Geographical Trends in Cerebrovascular Disease Mortality in the United States, 1999-2020*. Selected top 20 abstract for HEADS UP Symposium, International Stroke Conference; Los Angeles, CA; February 4, 2025.

Tran J. *RETFound Evaluation for Glaucoma Detection Using OCT Images*. Selected oral presentation for Medical Student Research Day; University of Nevada, Reno School of Medicine; November 20, 2024.

Pasia LG, **Tran J**, Flores D. *Trends and Disparities in Firearm-Related Mortality Among US Children and Young Adults, 1999-2020*. Top poster presentation at Inaugural Wolf Pack Pediatric Symposium; University of Nevada, Reno; May 23, 2024.

Service, Leadership, Experience

Ophthalmology Student Interest Group Leader Dec. 2024 - May 2025
University of Nevada, Reno School of Medicine *Reno, NV*

- Facilitate networking opportunities to learn about ophthalmology and connect with local ophthalmologists.

Street Medicine Student Outreach Clinic Manager Dec. 2024 - May 2025
University of Nevada, Reno School of Medicine *Reno, NV*

- Led the integration of the Street Medicine Student Interest Group into the established Student Outreach Clinic.
- Developed and implemented a structured schedule to ensure regular clinic operations.
- Successfully organized and operated 9+ clinics, reaching hundreds of homeless individuals. Facilitated medical care for dozens of patients and referred numerous individuals to the free Student Outreach Clinic.

Street Medicine Student Interest Group Leader Dec. 2024 - May 2025
University of Nevada, Reno School of Medicine *Reno, NV*

- Secured funding and donations to acquire essential supplies for individuals experiencing homelessness.
- Organized events where students assembled care kits for distribution to the homeless community.
- Facilitated partnerships with local organizations to maximize outreach and impact.

Life Skills Instructor Dec. 2023 - May 2025

Eddy House *Reno, NV*

- Teach classes on first aid, health education, and essential life skills to 50+ youth at risk for homelessness.
- Developed and implemented a structured schedule to ensure regular clinic operations.
- Develop and deliver engaging lessons on topics such as personal safety, hygiene, and emergency preparedness.

Projects

Curiosity Kids [HACKATHON FINALIST] | *VR Development, Lens Studio, Snap* Jun. 2026

- Created during AWE Reality Snap Hack 2026. Finalist among 20+ teams.
- Using Snap Spectacles and Lens Studio, created a child-centered curiosity engine that encourages children to interact and learn from the world around them.
- Helped with background research, wrote white paper, developed business plan, write documentation and presentation.
- Created front-end UI elements and backend identification workflow.
- Hackathon Devpost

Strabismus Project | *VR Development, C#* Mar. 2023

- Developed a virtual reality application to measure subjective vs. objective measures in strabismus patients.

Operating Suite X-ray Visualizer | *Augmented Reality, Hololens, C#* Jun. 2020

- Developed an augmented reality (AR) application using the Microsoft Hololens 1, visualizing scattered x-ray to improve education and awareness of scattered radiation in interventional suites. Resulted in a poster presentation.

Stroke Rehabilitation with Mobile AR | *Augmented Reality, Unity, C#, Microsoft Azure* Feb. 2019

- Developed a mobile augmented reality application designed to encourage stroke rehabilitation by guiding users through hand exercises.

Professional Membership

2025 – Present Member, BRAIN Core, University of Nevada, Reno.

2024 – Present Member, Association for Research in Vision and Ophthalmology (ARVO).

2023 – Present Member, American Heart Association (AHA).

Technical Skills

Languages: Python, C/C++, C#
Developer Tools: Git, Google Cloud Platform, VS Code, Visual Studio, Jupyter Notebook, Unity, Lens Studio

Personal Interests

Rock Climbing, Social Dance, Piano